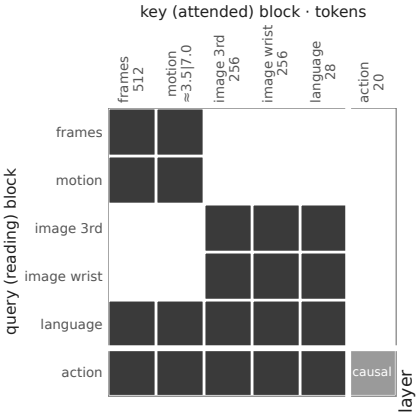
I. Two attention pathways carry motion-token content into the action tokens

Motion model, PatternLock | RouteStick, seed 42, 100 episodes / 495 inferences, head-averaged, episode-equal. (a) the attention mask as a block matrix and, per layer, where each block mainly reads; (b) the pathway edges per layer, both tasks pooled.

(a)

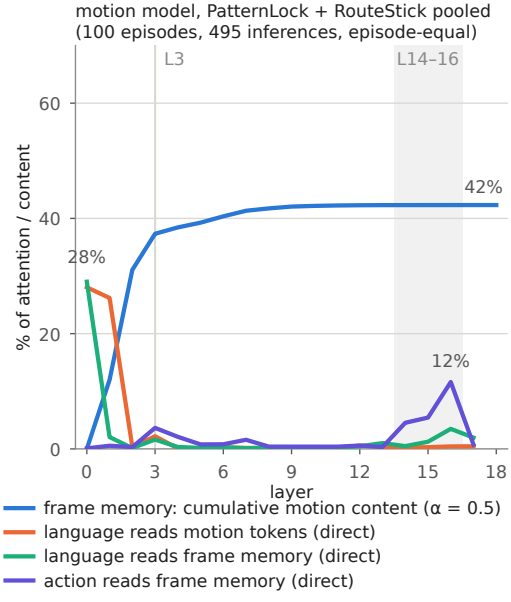


Block attention mask (dark = may attend; verified as exactly 0 elsewhere in every layer, head, episode). prefix = memory → images → language; suffix = action.

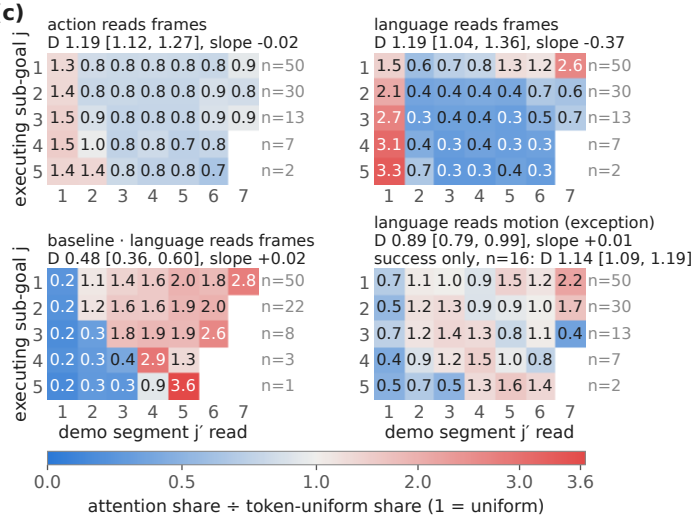
One row per layer. Each horizontal arrow runs from a source block to the block that reads it: the reader's largest external source at that layer (action: top two). Width and number = % of the reader's attention, printed beside the reader's column; colour = source. Two tasks pooled.

source: frame memory · source: motion tokens · source: current images · source: language

(b)



II. The pathways exist, but which demo segment is read shows no sub-goal alignment

RouteStick, motion model unless noted, 50 episodes per model, all inferences; rows pool episodes with different numbers of sub-goals m (n = episodes per row); frame stream weighted 'single', motion stream 'overlap'; D = diagonal enrichment, episode-level 95% CI.

(c)